

2-DAY INTENSIVE PROGRAMME

Mastering Excel, Microsoft Copilot & Power BI

for Modern Data Analysis and Interactive Reporting

Fakhrul Syahmi

Data Consultant & HRD Corp Certified Trainer

mufasyahcons@gmail.com



Excel



Copilot



Power BI

WELCOME

Your Trainer



Fakhru Syahmi

Data Consultant & HRD Corp
Certified Trainer



Academic

Master of Data Science (UKM). Research in bibliometrics, LDA topic modelling and machine learning.



Certified

HRD Corp Certified Trainer. Microsoft AI-900, DP-900 and PL-900.



Consulting

Power BI, Excel, Power Platform and AI productivity for GLCs, MNCs, government and universities.



Training

Corporate programmes for clients such as Perodua, Bank Rakyat, TNB, Teleperformance and HELP University.

By the end of two days, you will be able to...



Prepare data

Clean and structure messy business data in Excel and Power Query.



Analyse

Apply essential, lookup and dynamic array functions and PivotTables.



Use AI responsibly

Prompt Microsoft Copilot effectively and verify its output.



Model data

Design a star schema and write core DAX measures in Power BI.



Build dashboards

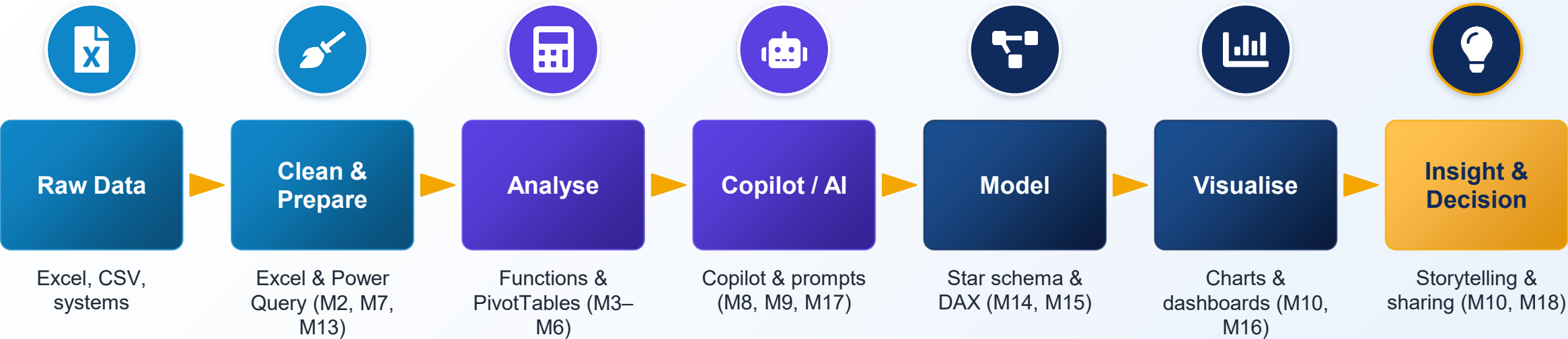
Create interactive reports that answer real business questions.



Communicate insight

Turn numbers into clear insights, actions and decisions.

One integrated workflow from raw data to decision



Day	Focus	Tools	Output
Day 1	Preparing and analysing data; AI-assisted analysis	Excel, Power Query, Copilot	Clean tables, analysis, Excel dashboard
Day 2	Modelling, DAX and interactive reporting	Power BI Desktop and Service	Published interactive report + capstone

Two days, nineteen modules, one capstone



DAY 1 · Excel & Microsoft Copilot

- 1 Modern Data Analysis in the AI Era
- 2 Preparing Business Data in Excel
- 3 Essential Excel Functions
- 4 Modern Lookup Functions
- 5 Dynamic Array Functions
- 6 PivotTables
- 7 Introduction to Power Query
- 8 Microsoft Copilot for Data Analysis
- 9 Prompt Engineering
- 10 Data Storytelling with Excel

Modules 1–10 · Excel, Power Query & Copilot hands-on



DAY 2 · Power BI for Interactive Reporting

- 11 Introduction to Power BI
- 12 Connecting Data
- 13 Data Transformation in Power Query
- 14 Data Modelling Fundamentals
- 15 Introduction to DAX
- 16 Visualisation & Report Design
- 17 Copilot in Power BI
- 18 Publishing, Sharing & Refresh
- 19 Capstone Case Study

Modules 11–19 · Modelling, DAX & capstone report

OUR CASE COMPANY

Maju Niaga Sdn Bhd

A Malaysian distributor of electronics, office furniture and office supplies selling to corporate, SME, government and dealer customers through five regions.



Training file	Used for
Day1_Excel_Practice_Dataset.xlsx	Day 1 exercises (Modules 2–10)
Monthly_Sales (18 CSV files)	Power Query and capstone
MajuNiaga_MasterData.xlsx	Dimension tables for Power BI
Sales_Targets_2026.xlsx	Unpivot practice and targets
Case Study Participant Brief	Exercises and capstone tasks

DAY 1

Excel & Microsoft Copilot for Data Analysis

- 1 Modern Data Analysis in the AI Era
- 2 Preparing Business Data in Excel
- 3 Essential Excel Functions
- 4 Modern Lookup Functions
- 5 Dynamic Array Functions
- 6 PivotTables for Rapid Analysis
- 7 Introduction to Power Query
- 8 Microsoft Copilot for Data Analysis
- 9 Prompt Engineering
- 10 Data Storytelling with Excel

From static reports to modern analytics

Traditional Reporting

- Manual copy-paste each period
- Static tables and PDFs
- Days to produce
- "What happened?"
- High error risk

Modern Analytics

- Connected once, refreshed
- Interactive dashboards
- Minutes to refresh
- "What happened, why, what next?"
- Repeatable, auditable steps



Excel is your workshop, Power BI is your showroom, and Copilot is the assistant who helps in both. You still decide what gets built.

Four types of analytics

01



Descriptive

What happened?

Total sales by region last quarter.

02



Diagnostic

Why did it happen?

Northern sales fell because two dealers stopped ordering.

03



Predictive

What is likely to happen?

Forecast Q4 demand from three years of seasonality.

04



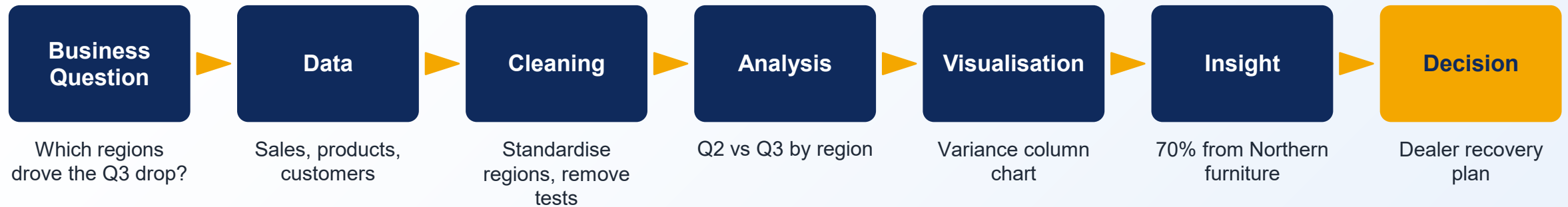
Prescriptive

What should we do?

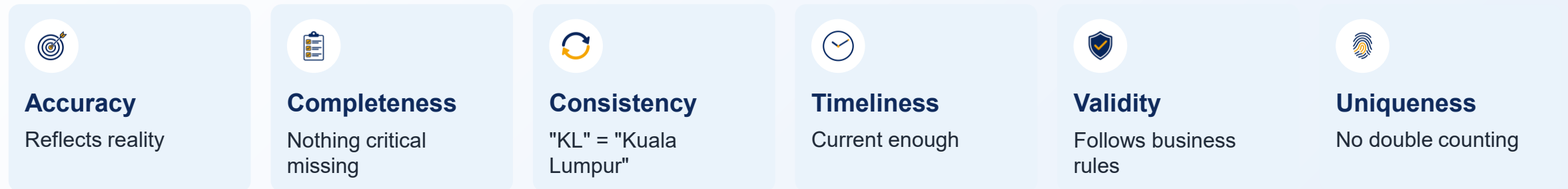
Reallocate stock to prevent stock-outs.

Increasing business value and analytical maturity

Start with the question, not the data



Six dimensions of data quality



Poor data leads to wrong decisions made with high confidence, which is worse than having no report.

What does analysis-ready data look like?

Typical branch extract

Date	Customer	Region	Amount
01/03/26	Syarikat ABC	KL	RM 4,500
2 Mar 2026	syarikat abc	Kuala Lumpur	4500
2026-03-02	Tech Hub	central	#N/A
	Tech Hub	Central	1,200
TOTAL			10,200

Analysis-ready table

OrderDate	InvoiceNo	Customer	Region	Amount
01/03/2026	INV-10231	Syarikat ABC	Central	4,500.00
02/03/2026	INV-10232	Tech Hub	Central	1,200.00

- One row per record, one column per field, one value per cell
- One header row with unique names; no merged cells, blank rows or totals
- Each column holds one data type; units in the header, not the cell
- A unique key (InvoiceNo) to detect true duplicates



Convert every dataset to an Excel Table with Ctrl + T and give it a name such as tblSales.

Common data issues and quick fixes

Issue	How to spot it	Fix
Numbers stored as text	Green triangle; SUM returns 0	Convert to Number, VALUE(), Text to Columns
Dates stored as text	ISNUMBER() returns FALSE	Text to Columns (DMY) or DATEVALUE()
Blank cells	Go To Special > Blanks	Fill, mark "Unknown" or remove
Duplicate records	Conditional Formatting > Duplicates	Data > Remove Duplicates on the key
Inconsistent spelling	Filter drop-down shows variants	Find & Replace (Ctrl + H); Data Validation
Extra spaces	LEN() longer than expected	TRIM(), then paste as values
Splitting / combining text	Several facts in one column	Text to Columns; Flash Fill (Ctrl + E)



Remove Duplicates is permanent. Work on a copy and choose key columns carefully.



HANDS-ON · MODULE 2

Clean the branch extract

Day1_Excel_Practice_Dataset.xlsx → Messy_Data sheet
Case Study Brief: Part A, 2.1–2.3

- 2.1 List every data quality problem on the Messy_Data sheet. How many data rows and duplicates are there?
- 2.2 Clean it into an Excel Table named tblBranchExtract: true dates, numeric amounts, standard regions, proper case, no TOTAL row.
- 2.3 Add a Region drop-down with Data Validation using tblRegion.
- ✓ Check your work with the Data Cleaning Checklist (handbook 2.6).

SUMIFS: the backbone of management reporting

```
=SUMIFS( tblSales[Amount], tblSales[Region], "Central", tblSales[Status], "<>Cancelled" )
```

1

Sum range first

The column to add up. (In SUMIF it comes last.)

2

Criteria pairs

Range + condition. Add as many pairs as you need.

3

Same size

All ranges must be the same size. Full table columns guarantee this.

Criteria type	Example	Meaning
Comparison	">10000"	Greater than 10,000
Not equal	"<>Cancelled"	Everything except Cancelled
Cell reference	">="&H2	Greater than or equal to H2
Wildcard	"*Sdn Bhd"	Contains "Sdn Bhd"

Function families you will use every week



Aggregate

SUM · AVERAGE · MIN · MAX · COUNT · COUNTA

`=AVERAGE(tblSales[Amount])`



Conditional

SUMIFS · COUNTIFS · AVERAGEIFS

`=COUNTIF(tblSales[Status], "Pending")`



Logical

IF · IFS · AND · OR · IFERROR

`=IF(AND([@Sales]>=[@Target],[@Att]>=95%),"Eligible","No")`



Text

LEFT · RIGHT · MID · LEN · TRIM · PROPER · TEXT · TEXTJOIN

`=MID("ELC-2026-0457",5,4) → 2026`



Date

TODAY · YEAR · MONTH · EOMONTH · DATEDIF

`=DATEDIF([@JoinDate],TODAY(),"Y")`



Rounding

ROUND · ROUNDUP · ROUNDDOWN

`=ROUNDUP([@Qty]/12,0) → cartons`

IFS: the first TRUE wins

```
=IFS( [@Score]>=90, "Excellent",  
      [@Score]>=75, "Good",  
      [@Score]>=60, "Satisfactory",  
      TRUE, "Needs Improvement" )
```

90–100 → Excellent

75–89 → Good

60–74 → Satisfactory

< 60 → Needs Improvement



Order matters

Check the highest threshold first; otherwise ">=60" catches everyone.



Add a catch-all

End with TRUE so every row gets a result instead of #N/A.



Use IFERROR wisely

Hide expected errors (divide by zero), not data problems you should fix.



HANDS-ON · MODULE 3

Answer business questions with formulas

Day1_Excel_Practice_Dataset.xlsx → *tblSales*,
tblEmployees
Case Study Brief: Part A, 3.1–3.7

- 3.1 Total sales excluding Cancelled (SUMIFS).
- 3.2 Central region sales in March 2026, excluding Cancelled.
- 3.3 Number of Pending order lines (COUNTIF).
- 3.4 Average Electronics line value (AVERAGEIFS).
- 3.5 Northern lines above RM10,000 (COUNTIFS).
- 3.6 Rating column with IFS: how many Excellent?
- 3.7 ServiceYears with DATEDIF to 30/06/2026: average for Active staff.

XLOOKUP: find it, return it

```
=XLOOKUP( lookup_value, lookup_array, return_array, [if_not_found], [match_mode], [search_mode] )
```

Task	Formula
Unit price for a product	=XLOOKUP([@ProductID], tblProducts[ProductID], tblProducts[UnitPrice])
Friendly message if missing	=XLOOKUP([@ProductID], tblProducts[ProductID], tblProducts[UnitPrice], "Not found")
Several columns at once	=XLOOKUP(H2, tblStaff[EmployeeID], tblStaff[[Name]:[SalesTarget]])
Latest order for a customer	=XLOOKUP(H2, tblSales[CustomerID], tblSales[OrderDate], , 0, -1)
Commission tier (approximate)	=XLOOKUP([@Sales2025], tblTier[MinSales], tblTier[Rate], , -1)



Sharing with Excel 2016/2019? XLOOKUP shows #NAME?. Use INDEX + MATCH: =INDEX(tblStaff[SalesTarget], MATCH(H2, tblStaff[EmployeeID], 0))

VLOOKUP vs XLOOKUP, and fixing #N/A

Feature	VLOOKUP	XLOOKUP
Default match	Approximate	Exact
Look to the left	No	Yes
Survives inserted columns	No	Yes
Built-in not-found message	No	Yes
Search from bottom	No	Yes
Available in	All versions	M365, Excel 2021+



Wrong value or column

Check both sides hold the same kind of ID.



Text vs number

"101" ≠ 101. Align types with VALUE() or TEXT().



Hidden spaces

TRIM() the IDs, then confirm the ID exists.



Duplicate IDs

First match is returned. Test with COUNTIF(>1).

One formula, many results: FILTER

```
=FILTER( tblSales,  
        (tblSales[Region]="Central") * (tblSales[Amount]>10000) ,  
        "No matching transactions" )
```

- * multiplies conditions = AND
- + adds conditions = OR

Spill range (outlined in blue in Excel)

OrderDate	Region	Product	Amount
03/01/2026	Central	Laptop Pro 14	22,500
07/01/2026	Central	Meeting Table	14,000
12/01/2026	Central	Workstation 4-Pax	27,200
...	...	...	...



#SPILL! means...

Cells in the way are not empty, the formula is inside a Table, or the range hits merged cells.



Refer to a whole spill range with #, e.g. =COUNTA(H4#). Requires Microsoft 365 or Excel 2021+.

Build a self-updating mini-report

Function	Example	Result
SORT	=SORT(tblSales, 7, -1)	All sales, largest Amount first
SORTBY	=SORTBY(tblStaff[Name], tblStaff[SalesTarget], -1)	Names ranked by target
UNIQUE	=SORT(UNIQUE(tblSales[Region]))	Alphabetical list of regions
SEQUENCE	=SEQUENCE(12)	Numbers 1–12 for months

```
H3: =SORT(UNIQUE(FILTER(tblSales[SalesRep], tblSales[Region]=H1)))
I3: =SUMIFS(tblSales[Amount], tblSales[Region], H1, tblSales[SalesRep], H3#)
K3: =TAKE(SORT(FILTER(tblSales, tblSales[Region]=H1), 7, -1), 5)
```

Reps in the region · their sales · top five transactions, all updating when H1 changes

Summarise thousands of rows in seconds

FILTERS
Year = 2026, Status ≠ Cancelled

COLUMNS
Category

ROWS
Region

VALUES
Sum of Amount

Show Values As	Answers
% of Grand Total	Share of each item
Running Total In	Year-to-date progress
% Difference From	Growth vs previous period

Questions a PivotTable answers

- Which region, product or rep generated the most revenue?
- How does monthly revenue trend against last year?
- What share of expenses does each department take?
- How many active employees per department?
- Which customers make up the top 20% of sales?

 Add Slicers and a Timeline, then use Report Connections to control several PivotTables at once.



HANDS-ON · MODULES 4–6

Look up, filter and summarise

Day1_Excel_Practice_Dataset.xlsx
Case Study Brief: Part A, 4.1–6.3

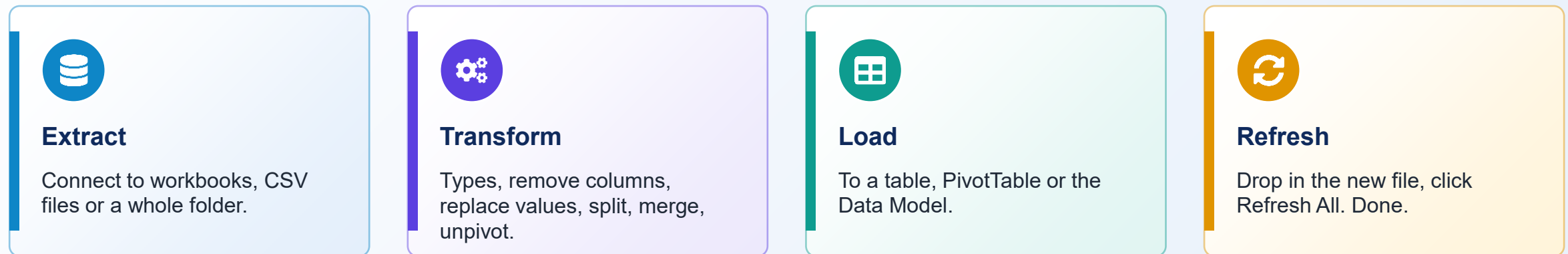
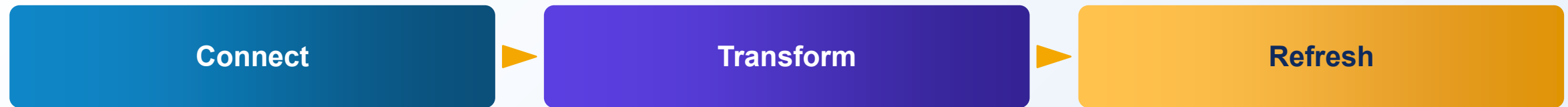
- 4.1 Add UnitCost with XLOOKUP and a LineCost column. Total LineCost excluding Cancelled?
- 4.3 Commission rate per rep with approximate-match XLOOKUP on tblTier. Total commission?
- 5.1 FILTER Central and Amount > RM10,000. How many rows spill?
- 5.2 Sorted unique list of East Coast customers. How many?
- 6.1 PivotTable: 2025 sales by region with % of Grand Total.
- 6.2 Monthly sales trend with Pivot Chart and Year slicer.
- 6.3 Active headcount by department.

Stop cleaning the same data every month

Traditional



Power Query



Append, Merge and Unpivot



Append = more rows

Stack January + February + March into one table. Same columns required.



Merge = more columns

Join sales to products on ProductID. Left Outer ≈ XLOOKUP; Left Anti finds records with no match.

Before: report layout

Department	Jan	Feb	Mar
Finance	50,000	48,000	52,000
HR	30,000	31,000	29,500



Unpivot Other
Columns

After: analysis layout

Department	Month	Budget
Finance	Jan	50,000
Finance	Feb	48,000
Finance	Mar	52,000
HR	Jan	30,000



HANDS-ON · MODULE 7

Combine a folder and unpivot a budget

*Monthly_Sales folder · Budget_Grid sheet
Case Study Brief: Part A, 7.1*

- 1 Data > Get Data > From Folder → Monthly_Sales. Combine & Transform.
- 2 Set types (use Locale for DD/MM/YYYY dates), trim and standardise Region.
- 3 Filter out Cancelled and C-TEST rows; Close & Load to a PivotTable.
- 7.1 Load Budget_Grid, Unpivot Other Columns. How many rows? Total 2026 budget?
- ★ Drop a copy of a CSV into the folder and click Refresh All. What happens?

Copilot: an AI assistant, not an autopilot

Capability in Excel	Example request	Always check
Generate formulas	"Add a column for profit margin"	Logic and zero handling
Explain formulas	"Explain column H step by step"	Matches your intent
Summarise data	"Sales by region and category"	Totals reconcile
Identify trends	"Highlight unusual months"	Business explanation
Highlight & sort	"Highlight orders above RM10,000"	Correct column
Charts & PivotTables	"Chart sales by region"	Right chart type
Draft insights	"3 bullets for management on Q1"	Accuracy, confidentiality



Prepare

Excel Table, clear headers, file in OneDrive or SharePoint.



Verify

Reconcile every figure with SUM, a filter or a PivotTable.



Features vary by Microsoft 365 subscription, Copilot licence, organisation configuration, tenant policies and app version. Check with IT.

Use AI responsibly



It can be wrong

Plausible formulas with incorrect logic, misread columns or overstated trends.



It lacks context

It does not know "exclude Void" or your KPI definitions unless you say so.



Confidentiality

Follow data classification; use only approved, enterprise-protected tools.



You are accountable

Reports you sign off are your responsibility, however they were produced.

Brief the AI like a new colleague

R

Role

"You are a financial analyst..."

C

Context

"...preparing the monthly management report."

T

Task

"Compare actual vs budget by department."

D

Data

"Use tblExpenses; exclude Status = Void."

O

Output

"A variance table plus five bullet findings."

C

Constraints

"Flag > 10%. Under 150 words. Use RM."

Weak

"Analyse this data."

Strong

"You are a business analyst. Analyse the monthly sales dataset by region and product category. Identify the top three growth areas, major declining segments and any unusual patterns. Present the findings in a management summary with supporting percentages."

Prompt library for everyday analysis

Purpose	Prompt
Data cleaning	"List data quality issues in this table (blanks, inconsistent spelling, duplicate invoice numbers, numbers stored as text) and suggest a fix for each."
Formula	"Write a formula returning total sales for the region in H1 and the month in H2 from tblSales using OrderDate and Amount."
Variance	"Compare Actual vs Budget by department for Q1. Show RM and % variance; list departments over budget by more than 10%, largest first."
Customers	"Group customers into High (> RM100,000), Medium and Low by total purchases; count each group and its share of revenue."
Management	"You are briefing the CEO. Write five bullets: performance vs target, top region, biggest risk and one recommended action. Use RM and %."



Iterate: "Exclude cancelled orders and recalculate" · "Show this by quarter" · "Explain how you calculated growth".

Data → Information → Insight → Action

Data

12,450 rows of sales transactions

Information

Q3 sales were RM4.2m, 12% lower than Q2

Insight

70% of the decline came from Northern furniture after two dealers stopped ordering

Action

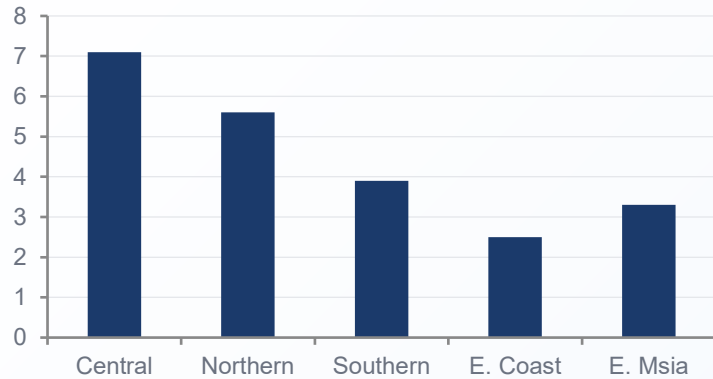
Assign a key account manager to recover both dealers by end of October



Write chart titles that state the insight: "Northern furniture sales fell 28% in Q3", not "Q3 Sales Chart".

Choose the chart for the message

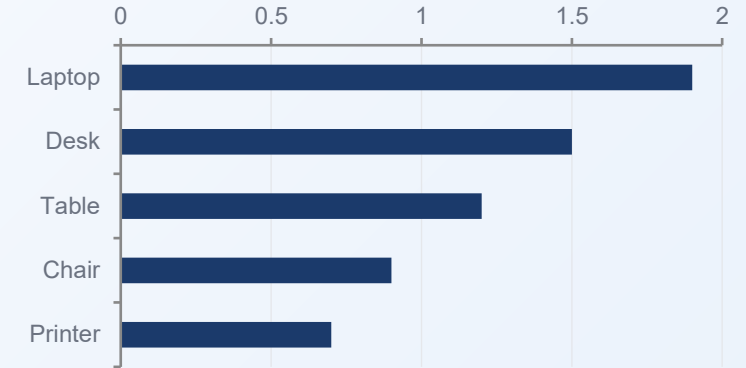
Comparison → Column



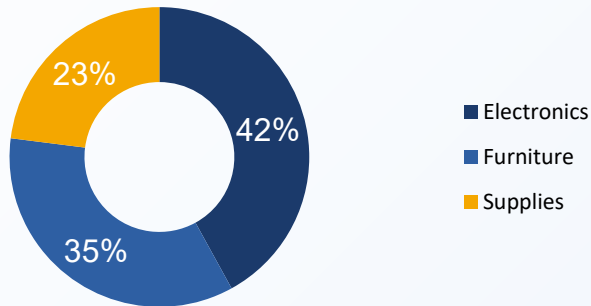
Trend → Line



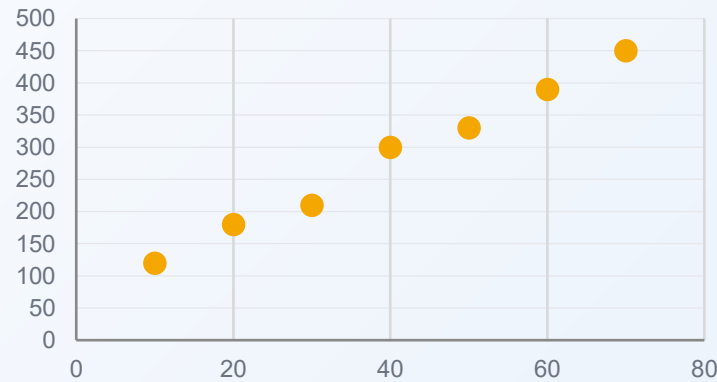
Ranking → Sorted bar



Share (2–4 parts) → Donut



Relationship → Scatter



Avoid

- 3D charts
- Pies with 5+ slices
- Lines for unrelated categories
- Truncated axes that exaggerate

Make what matters stand out



Conditional formatting

Highlight rules, Top/Bottom, data bars, colour scales, icon sets and formula rules such as `=H2="Overdue"`.



Sparklines

Tiny in-cell trend charts: 12-month sales next to each product total.



KPI indicators

Custom number format that shows ▲ in green and ▼ in red while the value stays numeric.

`[Color10]▲ 0.0%; [Red]▼ 0.0%;0.0%`

▲ 12.4% ▼ 8.1% 0.0%



Layout rule: key KPI top-left, one highlight colour for what matters, grey for context, units in every title (RM, RM '000, %).



HANDS-ON · MODULES 8–10

Copilot and a one-page dashboard

Day1_Excel_Practice_Dataset.xlsx → *tblExpenses*,
tblSales

Case Study Brief: Part A, 8.1–10.1

8.1

With Copilot (or manually), compare H1 2026 Actual vs Budget by department, excluding Void. Which department is most over budget? Verify with a PivotTable.

9.1

Write a ROLE–CONTEXT–TASK–DATA–OUTPUT–CONSTRAINTS prompt for a five-bullet Q2 2026 sales summary. Record the verified output.

10.1

Build a one-page Excel dashboard: three KPIs, a trend chart, a regional comparison and slicers. Give each chart an insight title.

DAY 1 RECAP

Clean data, the right function, a PivotTable and a well-briefed Copilot will answer most business questions.



Tables first

Ctrl + T and a good name



SUMIFS & XLOOKUP

Your everyday workhorses



Power Query

Connect · Transform · Refresh



Copilot

Brief clearly, always verify

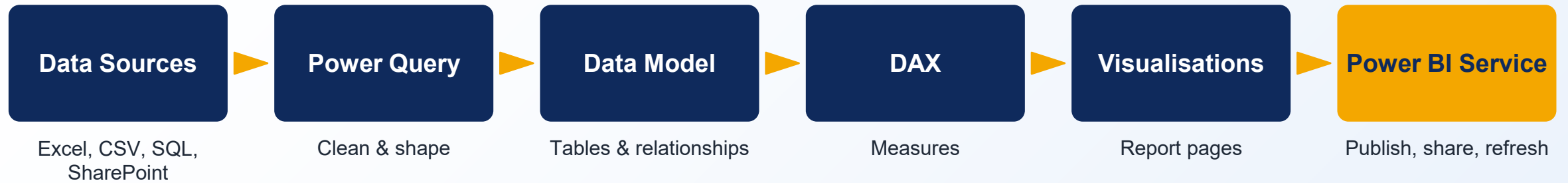
Tomorrow: the same data becomes an interactive Power BI report.

DAY 2

Power BI for Interactive Reporting

- 11 Introduction to Power BI
- 12 Connecting Data to Power BI
- 13 Data Transformation in Power Query
- 14 Data Modelling Fundamentals
- 15 Introduction to DAX
- 16 Visualisation & Report Design
- 17 Copilot in Power BI
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- 19 Capstone Case Study

How Power BI fits together



Power BI Desktop

Free Windows app to connect, transform, model and design reports.



Power BI Service

Cloud platform to publish, share, refresh and view on web or mobile.



Reports vs dashboards

Reports: multi-page, built in Desktop.
Dashboards: single page of pinned tiles in the Service.

Excel and Power BI: complementary, not competing

Criteria	Excel	Power BI
Dataset size	~1 million rows per sheet	Millions of rows, compressed
Analysis	Flexible cell-level, what-if, financial models	Consistent measures across the model
Automation	Power Query refresh, macros	Power Query + scheduled refresh
Dashboards	PivotTables and slicers	Purpose-built, highly interactive
Sharing	Files via email, OneDrive, SharePoint	Secure web/mobile with permissions and RLS
Interactivity	Slicers, timelines	Cross-filter, drill-down, drill-through, tooltips
Data modelling	Power Pivot (less visible)	Central, visual relationship design



Explore in Excel → publish shared reporting in Power BI → let users Analyze in Excel. Power Query and modelling skills transfer both ways.

Choose the right source and mode



Excel



CSV / Text



Folder



SQL Server



SharePoint



Web



Dynamics,
Dataverse, more

	Import	DirectQuery
How it works	Data copied into the model and compressed	Data stays in the source; queried live
Performance	Very fast	Depends on the source
Freshness	As of last refresh	Near real time
Typical use	Most business reports	Very large or operational databases

- Connect to the most authoritative source, not an edited copy
- Prefer structured tables over report layouts
- Store files in SharePoint/OneDrive so the Service can refresh
- Bring only the columns and rows you need

A standard cleaning workflow

01

Promote headers

02

Set data types

03

Remove unused columns

04

Rename columns

05

Handle nulls & errors

06

Standardise values

07

Remove duplicates

08

Add columns

09

Merge / Append

10

Rename query & apply

Custom column (M)

```
[Quantity] * [UnitPrice] * (1 - [Discount])
```

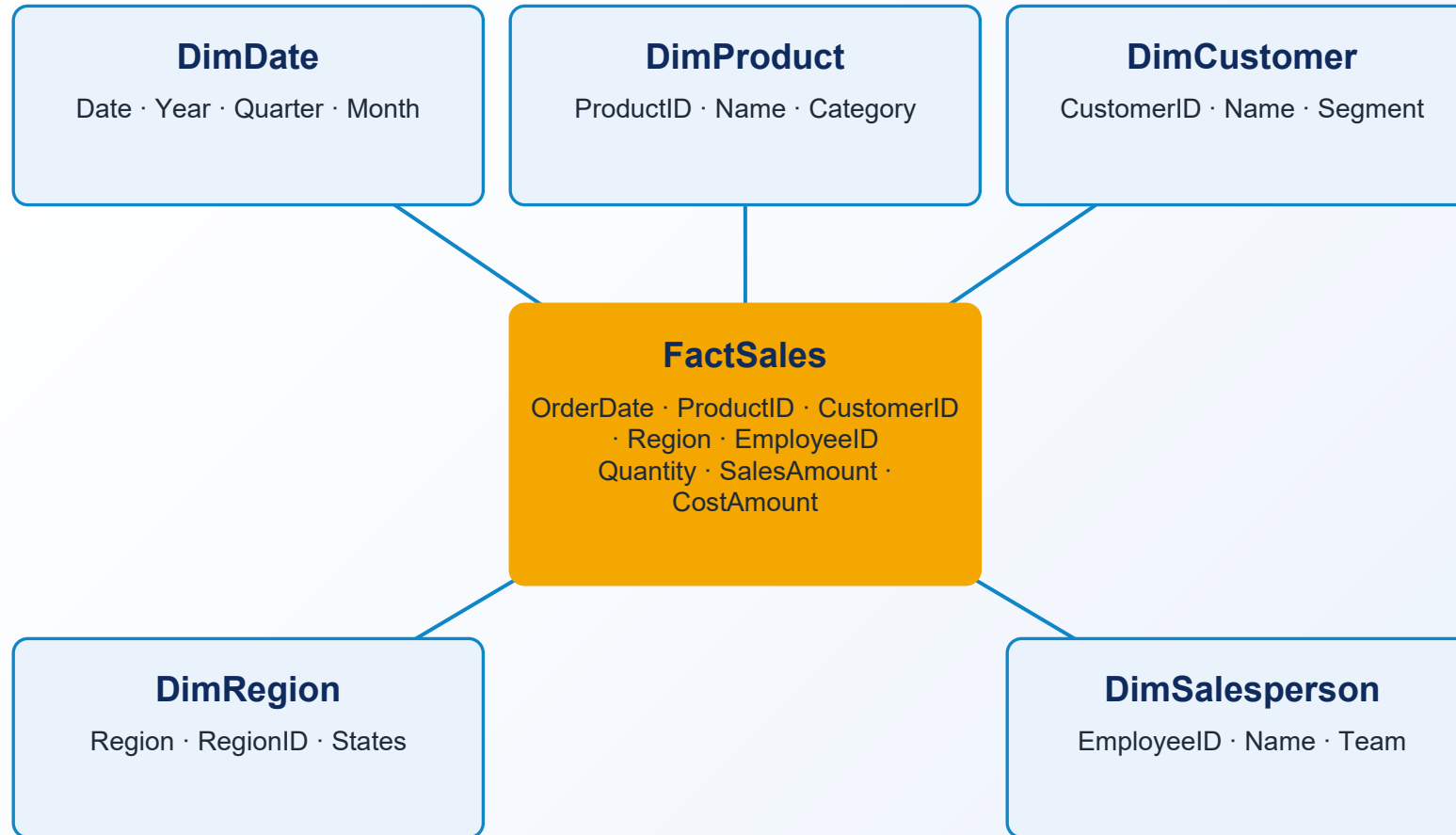
Conditional column (M)

```
if [SalesAmount] > 10000 then "Large"  
else if [SalesAmount] > 2000 then "Medium"  
else "Small"
```



Rename important steps and use Left Anti merge to find invalid codes.

Facts, dimensions and the star schema



*1 → * (one-to-many), single direction*

Fact table

Events and numbers to summarise.
Many rows. Foreign keys.

Dimension table

Who, what, where, when. Unique primary key.

Filters flow 1 → *

Selecting Furniture in DimProduct filters FactSales.

Every model needs a date table

```
DimDate      = CALENDAR( DATE(2025,1,1) , DATE(2026,12,31) )
Year         = YEAR( DimDate[Date] )
Month No     = MONTH( DimDate[Date] )
Month Name   = FORMAT( DimDate[Date], "MMM" )
Quarter      = "Q" & QUARTER( DimDate[Date] )
```

Mark as date table

Table tools → Mark as date table. Enables time intelligence.

Sort by column

Month Name sorted by Month No so Jan comes before Feb.

Avoid these modelling mistakes



Duplicate keys

Duplicate keys in a dimension create many-to-many relationships.



Bi-directional "fixes"

Switching direction to make a visual work causes ambiguity.



One flat table

Everything in one table is slow and hard to maintain.

Calculated column or measure?

Calculated Column

Calculated: Row by row, at refresh

Storage: Stored in the table

Best for: Slicers, axes, categories

```
Line Profit = FactSales[SalesAmount] -  
FactSales[CostAmount]
```

Measure

Calculated: On the fly, per filter

Storage: Not stored

Best for: Totals, ratios, KPIs

```
Total Sales = SUM(FactSales[SalesAmount])
```



If you need to slice by it or put it on an axis → column. If you need to add it up or calculate a ratio → measure.

Core business measures

Total Sales	= SUM(FactSales[SalesAmount])
Total Cost	= SUM(FactSales[CostAmount])
Profit	= [Total Sales] - [Total Cost]
Profit Margin	= DIVIDE([Profit], [Total Sales], 0)
Total Quantity	= SUM(FactSales[Quantity])
Average Sales	= AVERAGE(FactSales[SalesAmount])
Number of Customers	= DISTINCTCOUNT(FactSales[CustomerID])
Number of Lines	= COUNTROWS(FactSales)

Reuse measures

Profit builds on Total Sales and Total Cost; change once, update everywhere.

DIVIDE, not /

Handles divide-by-zero safely.

Name & format

Table[Column] for columns; [Measure] for measures. Format as RM or %.



Totals look wrong? Check relationships first. It is rarely the DAX.

CALCULATE and filter context

RM32.9m

Total Sales (no filter)

Region	Total Sales
Central	RM10.6m
Northern	RM8.2m
Southern	RM5.8m
...	...

Same measure, different answers: each row filters the calculation. That combination of filters is the filter context. (Illustrative figures.)

```
Central Sales    = CALCULATE( [Total Sales], DimRegion[Region] = "Central" )
All Region Sales = CALCULATE( [Total Sales], ALL( DimRegion ) )
Region Share %   = DIVIDE( [Total Sales], [All Region Sales] )
```

Time intelligence (needs a marked date table)

```
Sales YTD          = TOTALYTD( [Total Sales], DimDate[Date] )
Sales Last Year    = CALCULATE( [Total Sales], SAMEPERIODLASTYEAR( DimDate[Date] ) )
Sales YoY %        = DIVIDE( [Total Sales] - [Sales Last Year], [Sales Last Year] )
```

Anatomy of an executive dashboard



- One page, one purpose. Overview, regional, product.
- 6–8 visuals max for speed and clarity.
- Colour with meaning: reserve red/green for status.
- Insight titles and units (RM million).
- Accessibility: contrast, alt text, not colour alone.
- Test: can a manager answer 3 questions unaided?

Make the report interactive



Cross-filtering

Click a bar to filter the page. Adjust with Format → Edit interactions.



Drill-down

Year → Quarter → Month in one axis.



Drill-through

Right-click a customer to jump to its detail page.



Tooltips

Extra measures or a tooltip page on hover.



Conditional formatting

Colour margin below 10% in red.



Bookmarks & buttons

Switch views or reset filters.

Copilot in Power BI

Capability	Example prompt
Create report pages	"Create a page showing sales, profit and margin by region with a monthly trend."
Summarise	"Summarise key changes this quarter vs last quarter."
Answer questions	"Which category contributed most to the Northern decline?"
Assist with DAX	"Write a measure for year-over-year sales growth % using DimDate."
Document the model	"Explain what Region Share % calculates."



Make the model Copilot-ready

Business-friendly names, clean star schema, marked date table, hidden technical columns, measure descriptions and Q&A synonyms.

Validate every generated visual and measure against known totals.

 Typically requires a paid Microsoft Fabric capacity (F2 or higher) or Power BI Premium capacity, with Copilot enabled by the tenant admin. Availability also depends on region, licence and version, and changes often. Confirm with your Power BI administrator.

From Desktop to the organisation



Method	Best for	Consider
Workspace roles	Team members who build content	Admin / Member / Contributor / Viewer
Share link	A few colleagues	Recipients need licences
Apps	Departments or whole organisation	Easiest to manage at scale
Teams / SharePoint	Embedding where people work	Permissions still apply
Publish to web	Public data only	Anyone can view; often disabled

Scheduled refresh

Cloud sources refresh directly. On-premises sources need a data gateway.

Row-level security

Modeling → Manage roles, e.g. [Region] = "Central". Test with View as.

CAPSTONE

Maju Niaga Monthly Sales Performance

- Q1 H1 2026 vs H1 2025 by region
- Q2 Performance against 2026 targets
- Q3 Root cause in the weakest region
- Q4 Top products, categories, segments
- Q5 Sales reps vs target
- Q6 Growth and recommended actions

The scenario

The problem today

- Five branches email monthly ERP CSV files
- Finance spends ~3 days combining and cleaning
- Static charts ready only by the 10th
- H1 2026 looks weak, but nobody can say why

Memo from the Managing Director

"I want a report that refreshes itself, shows region, product, customer and salesperson performance against last year and our 2026 targets, and tells me clearly what is going wrong and what we should do."

18

monthly CSV files

5,500+

raw transaction lines

7

business rules to apply

4

master tables + targets

Business rules everyone must apply

1

Duplicates

Keep one copy per InvoiceLineID

2

Test rows

Remove CustomerID C-TEST

3

Cancelled

Exclude; Closed + Pending = sales

4

Invalid products

Exclude and list for Sales Ops (Left Anti)

5

Blank quantity

Replace with 0

6

Regions

Standardise to the five official names

7

Calculations

$\text{SalesAmount} = \text{Qty} \times \text{Price} \times (1 - \text{Discount})$

Never edit the CSV files by hand. July's file must refresh cleanly.

Nine tasks, one afternoon



Task 1

Frame the questions

10 min



Task 2

Combine & clean monthly files

45 min



Task 3

Master data & unpivot targets

20 min



Task 4

Explore in Excel with Copilot

20 min



Task 5

Build the star schema

20 min



Task 6

Write the measures

30 min



Task 7

Design the report

60 min



Task 8

Insights & recommendations

20 min



Task 9

Publish & present

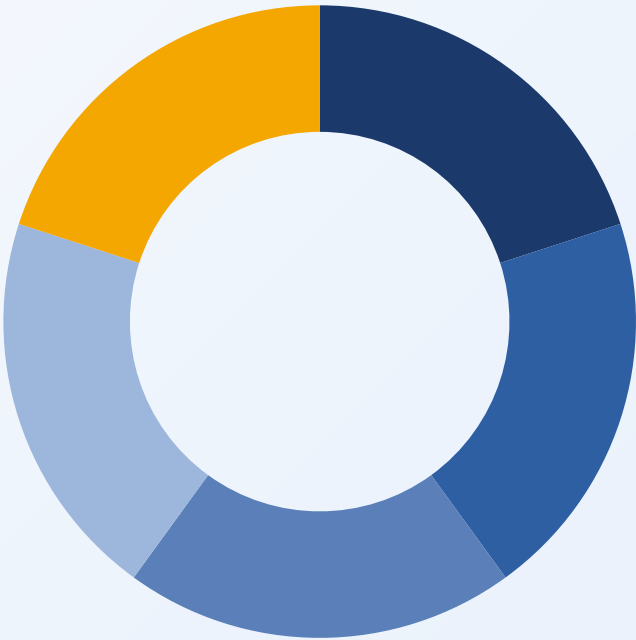
20 min



Detailed steps, the relationship table and all DAX measures are in the Case Study Participant Brief, Part B.

How your work will be assessed

Data preparation	All seven rules in Power Query; refresh works	20%
Data model	Star schema; 1 → * single direction; date table	20%
DAX measures	Correct, named, formatted, reused	20%
Report design	Clear layout, working interactions, insight titles	20%
Insight & recommendation	Data-backed answers; action, owner, date	20%



5 minutes to present to the "Management Committee"

Key takeaways

01

Question first

Every analysis starts with a specific business question.

02

Clean once, refresh forever

Power Query turns monthly effort into a click.

03

Model before visuals

A star schema and good measures make every visual correct.

04

AI accelerates, you verify

Brief Copilot clearly and reconcile its numbers.

05

Insight → action

Titles that tell, and recommendations with an owner and a date.

My 30-day action plan

One report to automate · three functions or measures to use · one Copilot prompt to adapt · one dashboard to build (handbook, Appendix A.4)

Thank you!

Questions, reflections and next steps



Fakhru Syahmi

Data Consultant & HRD Corp Certified Trainer

mufasyahcons@gmail.com